

Understanding Routing of SPM design using OpenLane

INTRODUCTION:

The Routing Lab in OpenLane focuses on creating physical interconnections between placed standard cells, macros, power grids, and clock networks. Routing is performed after Clock Tree Synthesis (CTS) and before signoff verification. The goal of routing is to connect all signal, clock, and power nets while satisfying design rule constraints (DRC), minimizing congestion, and ensuring signal integrity. OpenLane performs routing using open-source tools to generate detailed metal layer connections compliant with the selected PDK. Successful routing is critical for achieving a manufacturable and timing-clean ASIC design.

APPLICATION:

The Routing Lab in OpenLane is used to finalize interconnections for digital ASIC designs such as processors, controllers, and SoCs. It helps designers understand routing congestion, metal layer usage, via insertion, and DRC compliance. Routing is widely applied in educational ASIC flows, RISC-V cores, and open-source silicon implementations using Sky130. Proper routing ensures signal connectivity, power integrity, and readiness for physical verification and signoff.

OBJECTIVE:

After completing this lab, you will be able to:

Understand the role of routing in the RTL-to-GDSII flow

- Perform global and detailed routing using OpenLane
- Analyse routing congestion and metal layer utilization
- Verify signal, clock, and power connectivity
- Prepare the design for DRC, LVS, and signoff

PROCEDURE TO CREATE A LAB:

Step 1: Set up OpenLane flow

1.1: Invoke OpenLane in the terminal using the command: `openlane`

```
coe-4@ip-10-100-19-29:~/OpenLaneUser/designs$ openlane
=====
OpenLane started for user: coe-4
Workspace : /home/coe-4/OpenLaneUser
PDK Root  : /labroot/openlane_pdk
=====
OpenLane Container (ff5589f):/openlane$
```

1.2: Open folder- designs: `cd designs`

Step 2: Set up Project Directory

2.1: Create a directory for your lab: `mkdir <project_name>`

2.2: Open lab directory: `cd <project_name>`

2.3: Create directory for project: `mkdir spm`

2.4: Open project directory: `cd spm`

2.5: Inside project directory, create folder: `mkdir src`

2.6: Open folder: `cd src`

Step 3: Create Design file

3.1: Open vi editor for design with command: `vi spm.v`

Note: for RTL code refer the previous document of Demonstration of floorplan using openlane

Step 4: Create .json file

4.1: Open vi editor: `vi config.json`

```
{
    "DESIGN_NAME": "spm",
    "VERILOG_FILES": "dir::src/*.v",
    "CLOCK_PERIOD": 10,
    "CLOCK_PORT": "clk",
    "CLOCK_NET": "ref::$CLOCK_PORT",
    "FP_PDN_VOFFSET": 7,
    "FP_PDN_HOFFSET": 7,
    "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
    "FP_PDN_SKIPTRIM": true,
    "pdk::sky130*": {
        "FP_CORE_UTIL": 45,
        "scl::sky130_fd_sc_hd": {
```

```
"CLOCK_PERIOD": 10
},
"scl::sky130_fd_sc_hd11": {
    "CLOCK_PERIOD": 10
},
"scl::sky130_fd_sc_hs": {
    "CLOCK_PERIOD": 8
},
"scl::sky130_fd_sc_ls": {
    "CLOCK_PERIOD": 10,
    "MAX_FANOUT_CONSTRAINT": 5
},
"scl::sky130_fd_sc_ms": {
    "CLOCK_PERIOD": 10
}
},
"pdk::gf180mcu*": {
    "CLOCK_PERIOD": 24.0,
    "FP_CORE_UTIL": 40,
    "MAX_FANOUT_CONSTRAINT": 4,
    "PL_TARGET_DENSITY": 0.5
}
}
```

Step 6: Use the following commands to run the flow:

OpenLane to start the design flow in interactive mode

i) `flow.tcl -interactive`

Used in the OpenLane interactive shell to prepare your design for the digital IC design flow.

ii) `prep -design spm`

Runs yosys synthesis on the current design.

iii) `run_synthesis`

To initiate floorplanning

iv) run_floorplan

To initiate placement

v) run_placement

To initiate cts

v) run_cts

To initiate routing

v) run_routing

run_routing this command performs global routing followed by detailed routing, connecting all signal, clock, and power nets.

```
% run_routing
[STEP 14]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/14-resizer_design.log)...
[STEP 15]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/15-rsz_design_sta.log)...
[STEP 16]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/16-resizer_timing.log)...
[STEP 17]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/17-rsz_timing_sta.log)...
[STEP 18]
[INFO]: Running Global Routing (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/18-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 19]
[INFO]: Writing Verilog (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/18-global_write_netlist.log)...
[STEP 20]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/20-grt_sta.log)...
[STEP 21]
[INFO]: Running Fill Insertion (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/21-fill.log)...
[STEP 22]
[INFO]: Running Detailed Routing (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/22-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 23]
[INFO]: Checking Wire Lengths (log: designs/spm/runs/RUN_2026.01.29_06.44.46/logs/routing/23-wire_lengths.log)...
1769669517
%
```

Step 7: View plan using klayout:

Locate the DEF file & merged.nom.lef

i) designs/spm/runs/RUN_2026.01.23_05.16.33/results/routing/spm.def

ii) designs/spm/runs/RUN_2026.01.23_05.16.33/tmp/merged.nom.lef

Open in klayout in the same directory where spm.def is located

ii) klayout routing.def merged.nom.lef

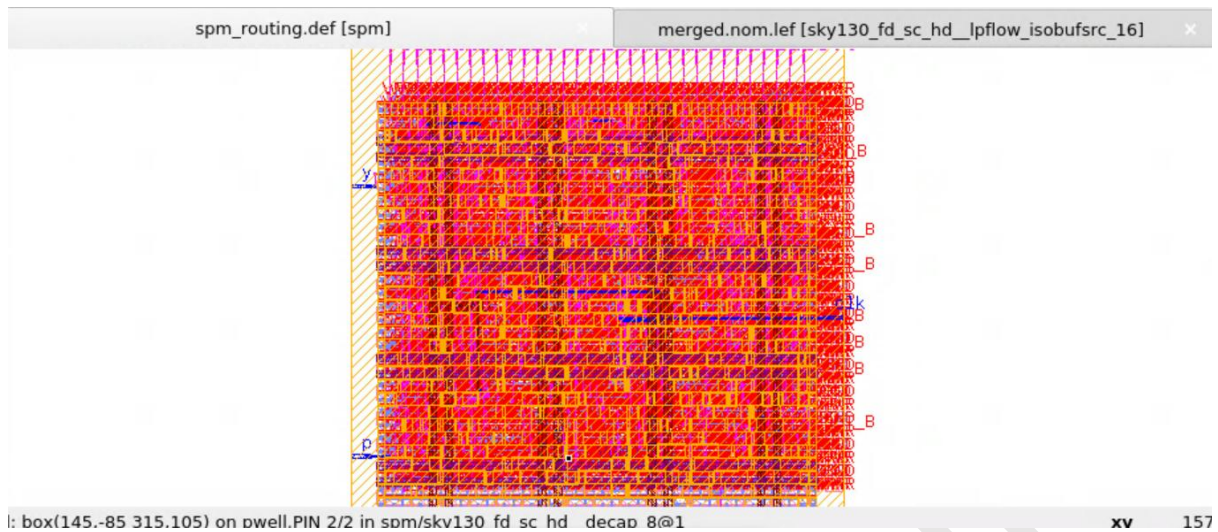


Figure 1: layout view of Routing.def

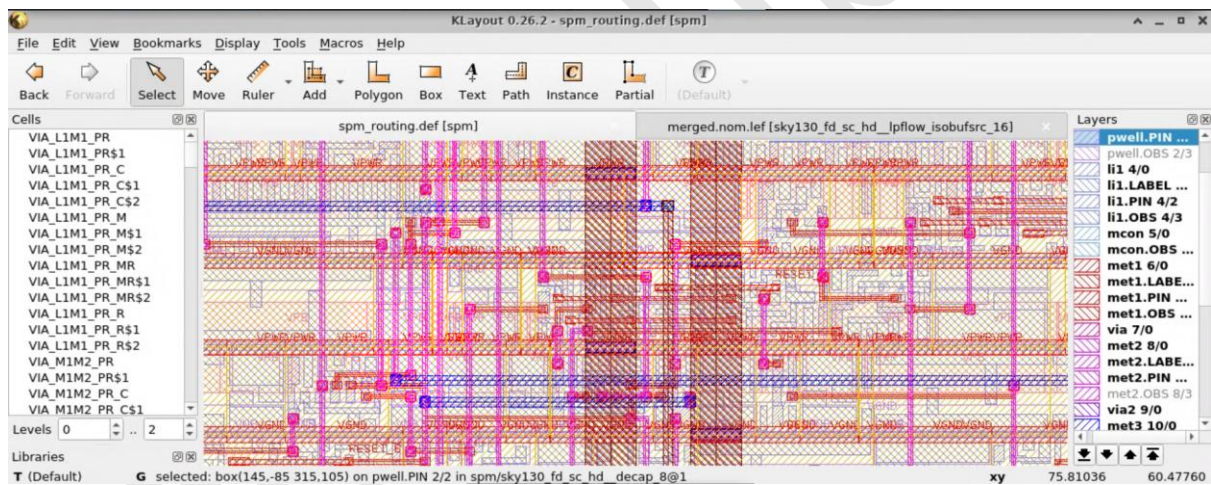


Figure 2: Zoomed in layout view of Routing.def

OBSERVATION:

- Successful routing of signal, clock, and power nets
- Metal layer usage across multiple routing layers
- Via insertion between metal layers
- Routed clock network from CTS stage
- No open nets or disconnected pins
- Minimal routing congestion

- Routed DEF visualization in klayout

OUTPUT:

Output of the Routing stage includes:

- spm.def file with complete routed layout
- Global and detailed routing reports
- Routed signal, clock, and power nets
- DRC-clean or near-clean routing results
- Routing-ready design for signoff checks

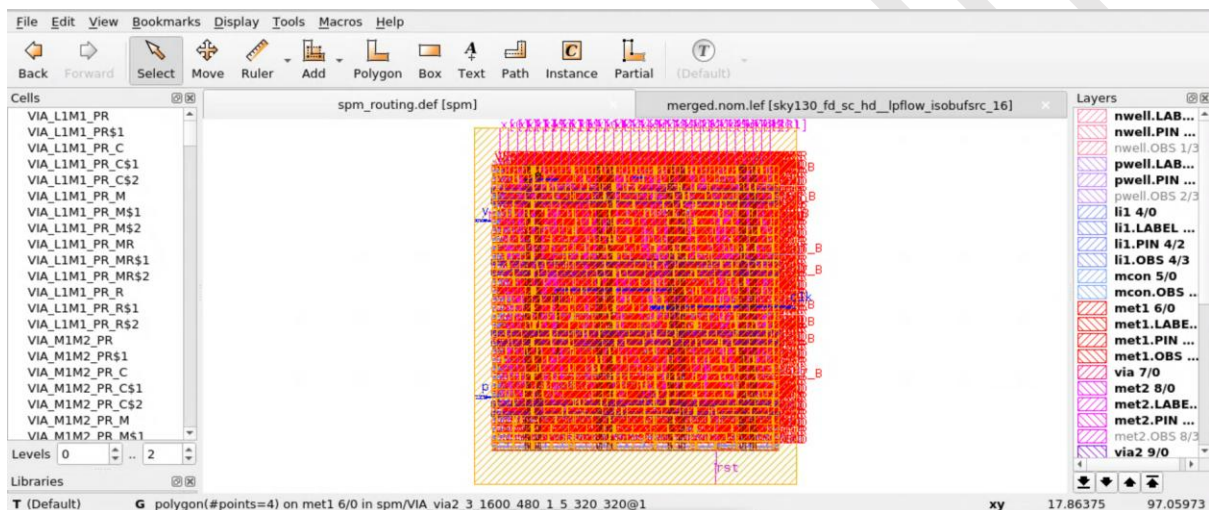


Figure 3: Output of Routing.def

CONCLUSION:

In this lab, OpenLane was successfully used to generate a fully routed layout for the given RTL design. The routing stage connected all signal, clock, and power nets while adhering to design rule constraints and metal layer guidelines of the selected PDK. The routed design was verified through reports and layout visualization using KLayout, confirming proper connectivity and routing quality. This successful routing implementation prepares the design for final physical verification steps such as DRC, LVS, and GDSII generation, ensuring manufacturability and reliable ASIC operation.